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Utilization of Absorbent Hydrophilic Polymer to Dry Water Based Drill Cuttings

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Abstract

Oil and Gas drilling rigs use waste pits on location for capturing drill cuttings and all liquid waste discharge. Closed loop drilling practices have been successfully demonstrated to greatly reduce the volume of liquid waste and separate from the water-based cuttings (WBC) using mechanical dryers' equipment. After extensive lab and field evaluation an Absorbent Hydrophilic Polymer (AHP) was used to effectively dry WBC using a method that is more economical than mechanical dryers.

Wide lab testing on multiple WBC were performed In-house in order to verify the absorbent properties of the hydrophilic polymer. WBC samples for different wells were collected and analyzed at the lab before and after the AHP treatment. Lab results showed absorption of the liquid phase with minimal AHP dose concentrations. Once the AHP performance was verified at the lab, we moved to the field evaluation. A candidate well was identified and 35 cubic meters of WBC were collected and treated. Key performance indicators (KPI's) were established as part of the evaluation criteria to assess the AHP performance.

A telehandler and bucket mixer were used on the field trial to estimate mixing time, assess the consistency of stabilized cuttings and determine the desired concentration of the AHP. The volume of cuttings treated (35 cubic meters) required an average AHP concentration of 0.62% (397.6 kg) and an average mixing time of 4 minutes. Additionally, an economical comparison between employing drying shakers and using AHP per well was done, taking into consideration that with drying shakers, all equipment and personnel costs are charged every day from rig up until rig down. With AHP utilization it is charged per used consumption meaning that the less cuttings come to surface, the less AHP is used and only while drilling. Thus, the cost reduces for a product that is dry, stackable and easily managed. In summary, after evaluating the collected data at the field and comparing with the established KPI's for the field evaluation, it was concluded that all KPI's were accomplished: field trial was conducted in a safe and controlled matter, usage was tracked and reported, WBC were stabilized and stackable after the treatment, no more than 20 kg per cubic meter were used and we observed full absorption of WBC's liquid phase.

This paper describes the utilization of Absorbent Hydrophilic Polymer to dry water-based cuttings and the excellent performance observed of this technology, contributing in the optimization of the waste management and minimization quest.

Introduction

Hydrocarbons trapped in subsurface reservoirs can be raised to the surface of the Earth (that is, produced) through wellbores formed from the surface to the subsurface reservoirs. Wellbore drilling systems are used to drill wellbores through a subterranean zone (for example, a formation, a portion of a formation or multiple formations) to the subsurface reservoir. At a high level, the wellbore drilling system includes a drill bit connected to an end of a drill string. The drill string is rotated and weight is applied on the drill bit to drill through the subterranean zone. Wellbore drilling fluid (also known as drilling mud) is flowed in a downhole direction through the drill string. The drilling fluid exits the drill bit through ports defined in the drill bit and flows through an annulus defined by an outer surface of the drill string and an inner wall of the wellbore. As the drilling fluid flows towards the surface, it carries any cuttings and debris released into the wellbore due to and during the drilling. The cuttings and debris are released from the subterranean zone as the drill bit breaks the rock while penetrating the subterranean zone. When mixed with the drilling fluid, the cuttings and debris form a solid slurry that flows to the surface. At the surface, the cuttings and debris are filtered and the wellbore drilling fluid can be recirculated into the wellbore to continue drilling. The cuttings and debris carried to the surface by the drilling fluid provide useful information, among other things, about the wellbore being formed and the drilling process.

In the Oil and gas drilling industry, a rig is used to drill by rotary and generate broken bits of solid material called drilled cuttings, which are removed from a borehole and are transported to the surface using a drilling fluid pumped downhole and returned to the surface pits. Those cuttings come to surface wet, essentially with drilling fluid.

Conventional drilling practices typically involve the creation of waste pits to store drilling cutting and excess drilling fluid, leading to potential soil contamination and other environmental hazards. This has led to growing interest in exploring more sustainable and environmentally friendly drilling practices. Additionally, the high consumption of drilling fluid in the conventional drilling methods leads to increased operational costs (Maryam, 2010). This issue is critical as the drilling industry strives to balance economic efficiency and environmental responsibility. In response to these concerns, the use of a closed loop system has emerged as an innovative solution. With its straightforward design and effectiveness, this system offers substantial environmental benefits and optimizes the use of drilling fluids. It consists of three main components: auger, secondary shaker and frack tank. The closed loop system operates by using the rig's existing power supply, thereby eliminating the need for an external power source. This energy-efficient approach enables satisfactory separation of cuttings, which can then be disposed by using skips boxes (Smith et.al 2022). Moreover, the system uses an air pump to recover clean mud, which can be reintroduced into the system. Any excess fluid is safely stored in the frack tanks and can be disposed of in a safe disposal area. In the study area waste pits were the historical method to capture and store cuttings and liquid waste on site. Site reclamation after drilling required these pits to be emptied and closed. This process often results in a considerable number of truckloads and generates huge volume of waste depending on the well profile and drilling time.

Closed loop drilling practices have been successfully demonstrated to greatly reduce the volume of liquid waste and separate the diesel contaminated cuttings from the water-based cuttings, leading to a substantial reduction in waste that must be trucked away. Drill cuttings exit the flowline from the Blow Out Preventors (BOP) riser and are first processed by the primary solids control equipment’s of the drilling rig, which are the shale shakers. These cuttings exit the primary solids control equipment with a liquid content of 30-50% water-based mud (WBM) or oil-based mud (OBM), depending the drilling fluid in use. These cuttings are too wet and soupy to be easily handled with front-end loaders or skip bins. In closed loop drilling they will be further processed with a drying shaker system which may consist of secondary shale shakers, a vertical dryer shale shaker, thermal cuttings dryer, or other liquid extraction equipment. This additional equipment may reduce the liquid content to 5-20% but also adds cost to the operation.

Some drilling rigs have been attempting closed loop drilling without the additional processing equipment. To handle the cuttings, they are mixed with sand to stabilize. This is increasing the cuttings volume by 3-4 times and makes the overall volume and weight of the waste stream unviable for hauling and disposal. To haul cuttings, they should be dry enough for simple handling and loading into containers without causing spills or leakage of contaminated fluid. Mixing agents are available to further dry the cuttings and stabilize them for safe and efficient handling. For this reason, we looked into the option to use cutting drying agents, based on the fact that performance mixing agents have the following features:

- Light weight such as polymer powder or compressed wood pellets. Low specific gravity prevents the mixing agent from adding significant additional weight to the cuttings which may reduce load size.
- Highly absorbent for WBC and oil-based cuttings (OBC). Some mixing agents are treated for specific applications: hydrophilic for WBC and hydrophobic for OBC.
- Specific application drying agents with superior performance are desirable. Some agents can be mixed at a ratio of agent to cuttings at less than 5%.
- Convenient packaging. Sacks and super-sacks as well as hopper and auger systems that make adding and mixing of the agent to the drill cuttings while in the 3-side bin are desirable.
- Low toxicity. Agents with inert chemical properties and non-hazardous classification are preferred.

Based on the above benefits, it was proposed to evaluate and add to the existing drilling operations acceptable drying agents for cuttings treatment on closed loop drilling operations. Drying agent was selected based on the performance with WBC and the performance criteria and evaluation methods were coordinated by specialized technical and operational team.

Experimental Evaluation

The objective of the experimental evaluation was to observe both WBC samples collected before and after the treatment with AHP, in order to ensure correct concentration of the chemical to get a baseline for the field evaluation.

The bench test was designed to simulate as closely as possible the environment on the rig to identify if mixing will be feasible at scale. Performance test was executed in the lab as per below:

The first part of the experimental evaluation consisted of the following:

- Measure specific gravity (density) of WBC samples, using mass balance.
- Measure % Oil, % Water, % Solids of WBC samples, using a retort.

For the second part of the experimental evaluation, we followed the below testing protocol:

- Started with 500 ml of WBC sample 1 in a beaker or appropriately sized vessel.
- Used a lab stand mixer to mix the stabilizer (set at lower speed)
- Added AHP until reaches a pasty consistency. Record the grams used to get pasty consistency.
- Reported if any heat is given off, any vapor is formed, any smell is present, and if the end product is solid and stackable.
- Repeated same procedure using WBC sample 2.

The results obtained for the first part of the evaluation, can be observed on Table 1:

Table 1—WBC Density and Retort Analysis

Properties	Unit	WBC Sample 1	WBC Sample 2
Density	SG	2.71	2.68
Oil by Retort	%	2	1
Water by Retort	%	33	37
Solids by Retort	%	65	62

The second part of the evaluation was performed as per established testing protocol as we can observe in Fig-1, where 500 ml beaker was used to measure WBC and AHP additions by weight were recorded for the treatment:

Figure 1—Measure of WBC and AHP Additions by Weight

Results on the second part of the evaluation, revealed that on WBC sample 1, 28 grams of AHP were necessary to reach a pasty consistency and on WBC sample 2, 30 grams were necessary, as we can observe below in Table 2:

Table 2—AHP Treatment Quantities

Additive	Unit	WBC Sample 1	WBC Sample 2
AHP Used	grams	28	30

It's worth to mention that no heat was given off, no vapor was formed, no smell was present and end product was solid as clay and stackable, as good example we can see on below Fig-2, WBC sample 2 before and after AHP treatment:

Figure 2—WBC Sample 2 Before and After AHP Treatment

A second hydrophilic additive with slightly different composition was tested at the lab. WBC samples were collected from the field and the same testing protocol was executed.

On below Fig-3, we can observe WBC sample 3 before and after AHP treatment:

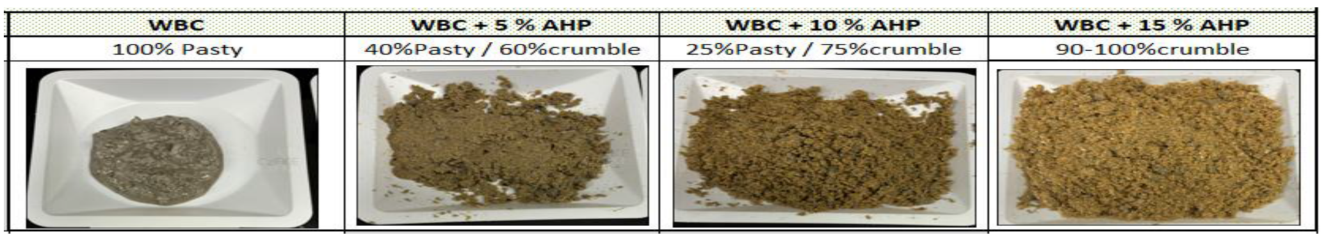


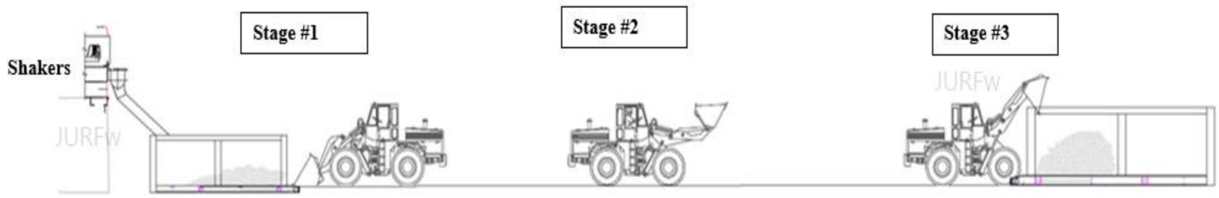
Figure 3—WBC Sample 3 Before and After AHP Treatment

Field Evaluation

Once the AHP performance was verified in the lab, we were ready to move to the next step, the field evaluation. Based on the lab resulted obtained, we agreed to use as AHP baseline concentration 20 kg per cubic meter. Key performance indicators (KPI's) were established as part of the evaluation criteria to assess the AHP performance. Proposed KPI's were:

- Field test must be conducted in a safe and controlled matter during treatment of 35 cubic meter (35 m^3) WBC.
- AHP usage must be tracked and WBC must be stabilized / stackable
- No more than 20 kg AHP per cubic meter should be used with full absorption of WBC liquid phase.
- Compliance with HSE & Environmental guidelines.

A candidate well was identified and we did a rig survey to evaluate all the conditions, potential impacts, type of hazards, risk controls measures in order to prepare the risk assessment. Then, we worked to get the game plan in place as we can see on below Fig-4:



Stage #1

Water-based cuttings from the shakers are discharged into a 3-sided bin through an open trough.

A telehandler equipped with a $0.5m^3$ capacity bucket (see Fig-6) and mixer scoops the cuttings from the 3-sided bin.

A WBC sample is taken for retort analysis to determine:

- Percentage of Water on Cuttings
- Density of the Cuttings

Stage #2

AHP is added to the cuttings inside the bucket.

The screw mixer inside the bucket is used to achieve a thorough mixture of cuttings and polymer until bonding is achieved.

A sample is taken for a second retort analysis.

Stage #3

Telehandler transfers and stacks the processed cuttings into a second 3-sided bin.

Figure 4—AHP Field Evaluation Game Plan

Following the above game plan, the wet WBC were collected as we can observe in below Fig-5:

Figure 5—Collected wet WBC

The treatment of the first three samples collected were done manually with the shovel while preparing the telehandler. This method was able to mix the AHP and within 30 minutes the WBC sample was properly treated until stackable conditions as we can observe below on Fig-6:

Figure 6—AHP Added Manual to WBC

Once telehandler was available, we used it to perform the treatment with the bucket mixer. The time of the treatment was reduced to 4 minutes. Below on Fig-7 you can see the telehandler and bucket mixer used on Stage # 1 and on Fig 8 the WBC after the AHP treatment.

Figure 7—Telehandler and Bucket Mixer Used on Stage # 1

Figure 8—WBC Before and After the AHP treatment

We were collecting samples for 10 days at location and following the agreed testing protocol. Also, we were recording data as part of the KPI's established in the evaluation criteria, See below on Table 4 recorded data:

Table 3—WBC Density and Retort Analysis

Properties	Unit	WBC Sample 3	WBC Sample 4
Density	SG	2.05	2.06
Oil by Retort	%	3	2
Water by Retort	%	35	36
Solids by Retort	%	62	61

Table 4—AHP Treatment Quantities

Test	Pre-Treatment Test		Post Treatment Test		Density Sample (S.G.)	Vol. of Cuttings Per Bucket (m^3)	No of Buckets Treated	Total Vol of Cuttings Processed (m^3)	Weight of Polymer Per Sample (kg)	Total Weight of Polymer Used (kg)	Mixing Time (Min) per sample	Remark
	% Water by vol.	% Solids by vol.	% Water by vol.	% Solids by vol.								
Test #1	60	40	44	56	1.7	1.2	3	1.2	4.9	14.7	30	Manual Mixing with Shovel
Test #2	62	38	42	58	1.7	1.2	3	1.2	7	21	30	Manual Mixing with Shovel
Test #3	56	44	42	58	1.7	1	1	1	7	7	30	Manual Mixing with Shovel
Test #4	64	36	44	56	1.7	0.5	3	1.5	9.1	27.3	4	Telehandler with bucket mixer
Test #5	62	38	42	58	2	0.5	3	1.5	6.3	18.9	3	Telehandler with bucket mixer
Test #6	60	40	40	60	1.8	0.5	3	1.5	4.9	14.7	3	Telehandler with bucket mixer
Test #7	54	46	28	72	1.7	0.5	18	9	4.2	75.6	3	Telehandler with bucket mixer
Test #8	54	46	40	60	1.7	0.5	8	4	4.9	39.2	3	Telehandler with bucket mixer
Test #9	60	40	40	60	1.7	0.5	10	5	9.1	91	5	Telehandler with bucket mixer
Test #10	40	60	26	74	1.9	0.5	18	9	4.9	88.2	3	Telehandler with bucket mixer

As we can observe on below Chart 1, the percentage of water before adding AHP was almost steady with minor differences, however the weight of AHP used per sample was fluctuating between 4.2 kg to 9.1 kg per cubic meter:

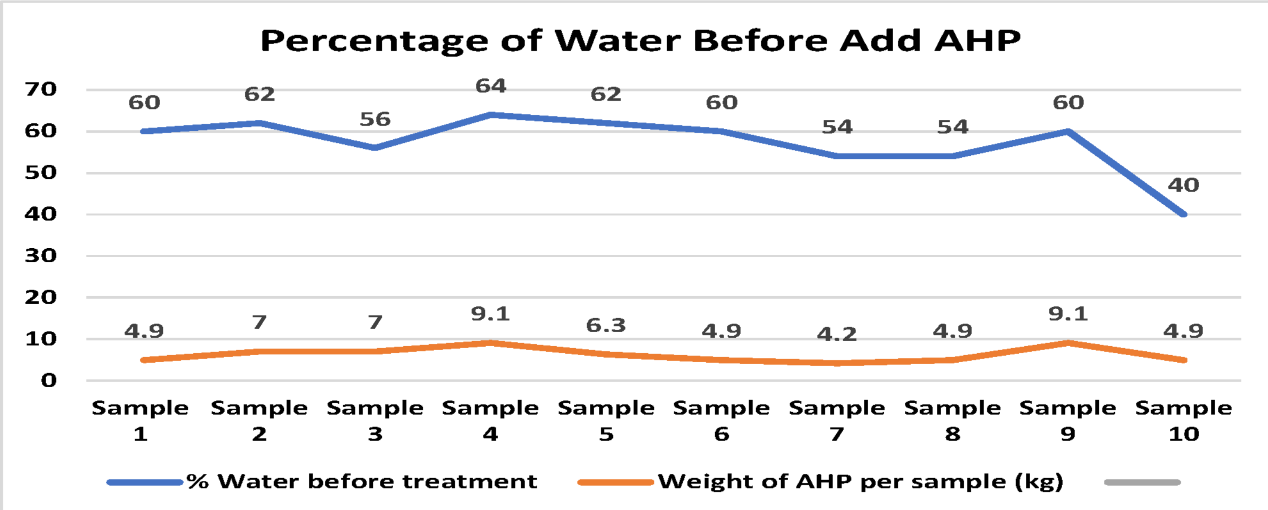


Chart 1—WBC Percentage of Water Before Adding AHP

Similarly, we ran retorts test on WBC before and after the treatment, as we can observe below on Table 5:

Table 5—WBC Retort Analysis Before and After AHP Treatment

Test	Sample Before Treatment							Sample After Treatment						
	Weight Empty Retort Cell (gr)	Weight Filled Retort Cell + Sample (gr)	Weight Graduated Cylinder - Empty(gr)	Wt. Retort Cell complete after retort cooled(gr)	Weight Graduated Cylinder + Condensate(gr)	Volume of Water in graduated cylinder(ml)	Volume of Sample(ml)	Weight Empty Retort Cell (gr)	Weight Filled Retort Cell + Sample (gr)	Weight Graduated Cylinder - Empty(gr)	Wt. Retort Cell complete after retort cooled(gr)	Weight Graduated Cylinder + Condensate(gr)	Volume of Water in graduated cylinder(ml)	Volume of Sample(ml)
Test #1	323	408	71.5	374.5	122.5	30.0	50	323	390	71.5	371.5	104	22.0	50
Test #2	323	408	71.5	374.5	124.2	31.0	50	323	380	71.5	358.5	96.6	21.0	50
Test #3	323	410	71.5	365	120.2	28.0	50	323	400	71.5	375	112	21.0	50
Test #4	323	423	71.5	401	129.5	32.0	50	323	397.5	71.5	385	124	22.0	50
Test #5	323	412	71.5	380	108.7	31.0	50	323	401.5	71.5	382.5	120	21.0	50
Test #6	323	412	71.5	382	107.1	30.0	50	323	403.5	71.5	387.5	109	20.0	50
Test #7	323	390	71.5	356.5	90.3	27.0	50	323	403.5	71.5	377	109	14.0	50
Test #8	323	401	71.5	374.5	107.4	27.0	50	323	403	71.5	381	115	20.0	50
Test #9	313	399	71.5	357	101	30.0	50	323	403.5	71.5	380.5	91.5	20.0	50
Test #10	835	930	71.5	906.5	90.5	20.0	50	636	695	71.5	682	84	13.0	50

Table 6—Analysis of Water on WBC and AHP Needed

% Water by vol.	% Solids by vol.	Weight of AHP per Sample (kg)	% by concentration of AHP
30	70	4	0.40%
35	65	5	0.47%
40	60	5	0.53%
45	55	6	0.60%
50	50	7	0.67%
55	45	7	0.73%
60	40	8	0.80%
65	35	9	0.87%
70	30	9	0.93%

Conclusions

The bucket mixer used on the field trial in conjunction with the telehandler was the most effective equipment to implement the AHP treatment compared with manual mixing and shovels. We were able to mix properly the AHP in average 4 minutes with the telehandler and bucket mixer.

For the volume of cuttings treated (35 cubic meters), required an average of AHP concentration of 0.62% (397.6 kg). An important benefit was shown that no additional waste volume increase, compared to some rigs having to mix sand with the cuttings to achieve a stable and stackable product which increases the total waste volume/weight 3-4 times. Additionally, an economical comparison between employing drying shakers and using AHP per well was done, taking into consideration that with drying shakers all equipment and personnel costs are charged every day from rig up until rig down. With AHP utilization, it is charged per used consumption meaning that the less cuttings come to surface, the less AHP is used, thus the cost reduces for a product that is dry, stackable and easily managed. In summary, after evaluating the collected data at the field and comparing with the established KPI's for the field evaluation, it was concluded that all KPI's were accomplished: field trial was conducted in a safe and controlled matter, usage was tracked and reported, WBC were stabilized and stackable after the treatment, no more than 20 kg per cubic meter (average 7 kg/m^3) were used and we observed full absorption of WBC's liquid phase.

Based on the trial and the experience gained from it, we can conclude that the below Table 6, represents the amount and percentage of AHP needed for WBC with respective percentages of water by volume.

As expected the higher the % of water by volume, the higher the AHP additions required to stabilize the WBC, however the concentration did not even reach the 1% concentration.

Nomenclature

WBC: Water-Based Cutting
AHP: Absorbent Hydrophilic Polymer
kg: Kilogram
m^3: Cubic meter
KPI: Key Performance Indicator
SG: Specific Gravity
ml: Milli-liter
OBC: Oil-Based Cutting
OBM: Oil-Based Mud
WBM: Water-Based Mud
BOP: Blow Out Preventors

gr: Grams
vol: Volume

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